CURRICULUM VITAE

Updated: 25.03.2026

Xiangtian Shi

Tel: +47 95553968
Email: xiangtian.shi@outlook.com
Oslo, Norway

Education

09.2026 – **MSc in Robotics, Systems and Control (Admitted)**
ETH Zurich, Switzerland

08.2025 – 06.2026 **BSc in Vehicle Engineering**
KTH Royal Institute of Technology, Sweden
- GPA: 4.9 / 5.0
- Relevant coursework: Automatic Control, Optimization, Applied Computer Science, Signals and Mechanical Systems, Electrical Engineering, Numerical Methods.

08.2023 – 06.2025 **BSc in Mechanical Engineering**
KTH Royal Institute of Technology, Sweden
- Completed foundational courses in mathematics, mechanics, and engineering.
- Transferred into the third year of the Vehicle Engineering programme after two years of study; all course credits fully recognized in the new programme.

Research & Technical Projects

01.2026 – 06.2026 **Physics-Informed Neural Networks for Optimal Path Planning of Unicycle Robots**
Bachelor Thesis, KTH Royal Institute of Technology, Sweden
- Formulating optimal path planning problems for nonholonomic unicycle robot models under kinematic and control constraints.
- Developing PINN frameworks that embed system dynamics directly into the learning process.
- Investigating constrained trajectory optimization with neural solvers.
- Implementing and evaluating models using PyTorch and numerical optimization methods.

01.2025 – 09.2025 **Rocket Avionics Engineer**
AESIR Student Rocketry Association, KTH Royal Institute of Technology, Sweden
- Developed numerical algorithms for apogee prediction, improving accuracy from 10% to 1.3% in RocketPy reference simulations.
- Designed and tuned a PID controller for active stabilization, achieving altitude control within 2.7% error at 3000 m.

01.2024 – 06.2024 **Solar-Powered Model Vehicle Project**
Project Course, KTH Royal Institute of Technology, Sweden
- Designed and built a solar-powered model vehicle, integrating mechanical and electrical subsystems.
- Implemented system control using Arduino (C++) and conducted performance testing.

Professional Experience

09.2025 – 06.2026 **Teaching Assistant in Electrical Engineering**
Unit of Mechatronics, KTH Royal Institute of Technology, Sweden
- Supporting laboratory sessions in circuit theory, signal processing, and mechatronics.
- Evaluating student assignments and assisting in course coordination.

05.2023 – 01.2025 **Mathematics Tutor**
MentorNorge AS, Norway
- Provided individualized tutoring in mathematics.

10.2022 – 06.2023 **Kindergarten Assistant**
Høvik kindergarten, Norway
- Supported children aged 2–4 in daily routines and guided them during creative and educational activities.

Technical Skills

Programming	Python, MATLAB, Arduino (C++)
Engineering Tools	PyTorch, Git, Solid Edge, ANSYS

Languages

Norwegian	Native
Chinese	Native
English	Fluent; IELTS Academic: 7.5
Swedish	Advanced; level B2

Extracurricular Activities

09.2023 – 06.2024 **KTH Student Council, International Group**
- Organized social and cultural events to enhance integration and collaboration among international students.

2019 – 2022 **Competitive Esports**
- Champions of PUBG Mobile Pro League Western Europe 2021 Spring Season, among 16 professional teams.
- Competed internationally in team-based esports tournaments.

Hobbies & Interests

Violin	10+ years of practice, previously a regional competition winner.
Table tennis	Active within the regional table tennis club and participated in regional competitions.
Other	Hiking, Jogging (Stockholm Half Marathon 2025), Skiing.